

W

LV.0 → LV.1 · AI ENGINEER

5 W-words that move you from Level 0 → 1.

Weights, Word2Vec, World Model & more. **Plain-English** definitions, plus the W-named tools, channels, and minds to follow.



↓ UP NEXT • TERMS

The W stack.

Weights

Weights & Biases

Word2Vec

World Model

Warehousing

W

Weights

learnable numbers • the model's memory

The internal numerical parameters a neural network adjusts during training, **weights are literally what a model learns**: adjust them correctly and the model solves the task.

USED IN

- Training
- Backpropagation
- Inference



Drop diagram • Weights

NEURONS • CONNECTIONS • VALUES

Weights & Biases

weights scale inputs • biases shift output

The two core parameters of a neural network:
weights set the importance of each input, biases shift the output independently.

USED IN

- Neural networks
- Training
- Decision boundaries



Drop diagram • Weights and Biases

$$Y = W \cdot X + B$$

WORD2VEC • LANGUAGE

Word2Vec

words as vectors • context as meaning

Google's landmark technique that maps words to dense vectors based on context, showing that **vector arithmetic captures semantic meaning**: king - man + woman = queen.

USED IN

- NLP embeddings
- Semantic search
- Similarity



Drop diagram • Word2Vec

KING - MAN + WOMAN = QUEEN

WORLD MODEL • AGENTS

World Model

internal simulation • predict before acting

An internal representation an agent builds to **simulate and predict how the world responds to actions**, enabling planning without real-world trial and error. Central to LeCun's vision for next-gen AI.

USED IN

- Autonomous agents
- Reinforcement learning
- Planning



Drop diagram • World Model

PERCEIVE → MODEL → PLAN → ACT

DATA WAREHOUSING • INFRASTRUCTURE

Warehousing

collect • store • query at scale

The process of centralizing large volumes of structured data for analytics and ML, **a data warehouse is the source of truth** that feature stores, dashboards, and AI pipelines all pull from.

USED IN

- Analytics
- Feature stores
- AI pipelines



Drop diagram • Data Warehousing

SOURCES → WAREHOUSE → QUERIES

↓ UP NEXT • TOOLS

The W stack.

Four W-named tools the AI stack runs on, from **experiment tracking and speech recognition** to vector search and computation.

W

MLOPS PLATFORM

Weights & Biases

experiment tracking • model evaluation



Drop W&B logo

The leading MLOps platform for **experiment tracking, hyperparameter tuning, model evaluation, and dataset versioning**, now standard in research labs and production ML teams alike.

Tracking

Every run, metric, and artifact

W&B

Shorthand for the platform

Sweeps

Automated hyperparameter search

WHAT IT IS

ML experiment and model tracking.

IN AI STACKS

Training logging, eval, and artifact store.

WHY DEVS USE IT

Understand what every training run did.

FIND THEM

wandb.ai

SPEECH RECOGNITION MODEL

Whisper

OpenAI • speech to text • 99 languages



Drop Whisper / OpenAI logo

OpenAI's open-source automatic speech recognition model that **transcribes and translates speech across 99 languages** with high accuracy, widely used in transcription, voice AI, and accessibility tools.

99

Languages supported

Open

Fully open-source weights

680K hrs

Trained on labeled audio

WHAT IT IS

Multilingual speech-to-text model.

IN AI STACKS

Voice input, meeting notes, accessibility.

WHY DEVS USE IT

Best open transcription available.

FIND THEM

openai.com/research/whisper

COMPUTATIONAL KNOWLEDGE ENGINE

Wolfram Alpha

compute • answer • reason exactly



Drop Wolfram Alpha logo

A knowledge and computation engine that uses symbolic math and curated data to **answer questions with exact, citable answers**, widely used alongside AI for reliable computation and factual grounding.

Exact

Symbolic computation, not guessing

2009

Founded by Stephen Wolfram

Tool

Plugged into GPT and agents

WHAT IT IS

Symbolic computation and knowledge engine.

IN AI STACKS

Tool-use layer for math and facts.

WHY DEVS USE IT

Exact answers where LLMs hallucinate.

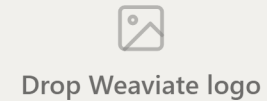
FIND THEM

wolframalpha.com

OPEN-SOURCE VECTOR DATABASE

Weaviate

vector search • RAG • open-source



An open-source vector database built for **semantic search, RAG, and recommendation systems**, with built-in vectorization modules and hybrid keyword plus vector retrieval.

Open

Open-source core, cloud option

Hybrid

Vector + keyword search together

Modules

Built-in vectorization at ingest

WHAT IT IS

Open-source vector DB for AI apps.

IN AI STACKS

RAG retrieval and semantic search layer.

WHY DEVS USE IT

Flexible hybrid search, open license.

FIND THEM

weaviate.io

↓ UP NEXT • CHANNELS TO FOLLOW

Learn from the best.

Two W-channels worth subscribing to, one for **visual math and ML storytelling**, one for deep dives into computation and AI from Stephen Wolfram himself.



YOUTUBE CHANNEL

Welch Labs

@WelchLabsVideo • visual ML storytelling



Drop Welch Labs logo

Stephen Welch's channel making **complex ML and math intuitive through high-quality visual storytelling**, especially praised for neural network and machine learning explainer series.

Visual

High-production explainer videos

Math

Intuition-first, then formulas

Free

Full series on YouTube

BEST FOR

Intuitive ML and math from scratch.

COVERS

Neural networks, ML, complex math.

TEACHING STYLE

Visual-first, story-driven, precise.

FIND THEM

youtube.com/@WelchLabsVideo

YOUTUBE CHANNEL

Wolfram

@WolframResearch • computation + AI



Drop Wolfram Research logo

Stephen Wolfram's educational channel exploring **mathematics, AI, and computational thinking** through live streams, talks, and deep-dive technical content on the nature of intelligence.

Wolfram

By Stephen Wolfram himself

Live

Streams and deep-dive talks

Depth

Math, AI, and computation

BEST FOR

Computational and mathematical depth.

COVERS

Computation, math, AI, physics.

TEACHING STYLE

Technical, curious, long-form.

FIND THEM

youtube.com/@WolframResearch

↓ UP NEXT • PEOPLE TO FOLLOW

The W minds.

Two W-named builders worth following: an **OpenAI co-founder** who helped invent Codex and GPT, and the CTO who built the cloud infrastructure AI runs on.

W

OPENAI CO-FOUNDER

Wojciech Zaremba

OpenAI • @wojzaremba



Co-founder of OpenAI working across robotics, RL, and LLMs, he **played a key role in building Codex and GitHub Copilot** and continues to lead core research at OpenAI.

OpenAI

Co-founder since 2015

Codex

Led the Codex and Copilot work

Robotics

RL and dexterous manipulation

KNOWN FOR

Codex, Copilot, and early LLM work.

BEST FOR

Code AI and frontier LLM thinking.

WHY FOLLOW

Inside view on OpenAI research.

FIND THEM

x.com/wojzaremba

CTO OF AMAZON

Werner Vogels

Amazon • @Werner



CTO of Amazon and the chief architect of AWS, the cloud infrastructure **that powers most of the world's AI training and serving**. A pioneer of distributed systems at scale.

Amazon

CTO since 2005

AWS

Helped build the AI cloud layer

Scale

Distributed systems pioneer

KNOWN FOR

AWS and large-scale distributed systems.

BEST FOR

MLOps, cloud, and infrastructure thinking.

WHY FOLLOW

Cloud infrastructure and AI at scale.

FIND THEM

allthingsdistributed.com

The **W** stack, in one card.

Save it. Drop a comment with the W-term you use most.

01	Weights	foundations	02	Weights & Biases	foundations
03	Word2Vec	language	04	World Model	agents
05	Warehousing	infra	+	Weights & Biases • Whisper	tools
+	Wolfram Alpha • Weaviate	tools	►	Welch Labs • Wolfram Research	channels
@	Wojciech Zaremba • Werner Vogels	people			



FOLLOW FOR THE FULL A TO Z
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X Next